

Examiner reports

Q1.

Drawing the force diagram in part (a) did cause difficulties for the candidates with several wanting to include a force directed down the slope. Part (b) was done correctly by almost all of the candidates. In part (c), several candidates did not form a three term equation, either omitting the component of the weight down the slope or the friction force. Most candidates produced a good attempt at part (d).

Q2.

Part (a) was either done very well or very badly. Some candidates produced simple, correct arguments. Two common incorrect approaches were to take $v = V \sin \alpha$ instead of $u = V \sin \alpha$, or to omit the negative sign from the acceleration when using the equation as $v^2 = u^2 + 2as$. The other approach that caused difficulty was when candidates tried to find

the time for the maximum height by working with the expression $y = V \sin \alpha - \frac{1}{2}gt^2$. This led to a variety of problems. The candidates who used the formula from part (a) usually had no difficulty finding V . A number of candidates did not use this result and often ended up with the vertical component of the initial velocity instead of the required speed. Part (c) caused problems for some candidates, the biggest problem being finding the time of flight correctly. Some used an equation that would have given the time for the maximum height by considering $y = 6$.

Q3.

Candidates found part (a) very straightforward, but had more difficulty with parts (b) and (c). Here, some did not know how to proceed and so did not make any real attempt at these parts of the question. This was particularly true of part (c). In part (b), the most common incorrect approach was to use the equation $\mathbf{r} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$, but taking $\mathbf{v}$ to be the velocity at $t = 2$ and not $t = 3$. A significant number of candidates left their answer as a vector and did not find its magnitude. There were very few correct solutions to part (c), with many candidates not making any attempt at this part of the question. Of those that did, very few realised that they needed to work with the magnitude of the velocity rather than the velocity vector. Several candidates produced equations where 10 was put equal to a vector.

Q4.

The first three parts of this question were generally done very well. Some candidates made arithmetic or algebraic errors, but these were not common place. In part (d), a few candidates simply calculated the product of the mass and the acceleration.

Q5.

Part (a) was done very well, but part (b) proved to be more demanding with only a few candidates gaining many of the marks available on this part of the question. Most candidates failed to set up the quadratic equations needed to solve the problem. A few candidates tried to apply the quadratic equation formula to find t , but taking the

coefficients as vectors, that is $a = -i + 2j$, $b = 4i + 7j$ and $c = -3i - 5j$. The candidates who realised how to approach the problem often gained full marks, although a few did run into difficulties having first stated the correct quadratic equations.

Q6.

Part (a) was done very well and many candidates were able to obtain the acceleration given in the question from correct working. Part (b) proved to be more problematic. The equation of motion was often set up incorrectly, the negative acceleration being a particular problem for some. In part (b), a number of candidates subtracted the 300 N from their previous answer rather than adding it on. Some candidates did not realise that they could simply modify their answer to part (b) and did a lot more work than was needed. There were some good answers to part (d), often expressed well. There were also a number of candidates who gave spurious answers.

Q7.

Parts (a) and (b) were done well by the vast majority of candidates. Most of the candidates who lost marks here had made minor arithmetic errors. Part (c) was a little more challenging, although there were many correct solutions. The main error made by the candidates was to use 100 m instead of $(100 - 16)$ m.

Q8.

Part (a) of this question was generally done very well, with many correct solutions. It was good to see a large number of candidates setting up an equation of motion for each particle and then using these to obtain the acceleration. The candidates seemed familiar with the topic and knew exactly what the question required.

The candidates found part (b) very much more demanding and there were relatively few correct solutions. Some candidates did not attempt this part of the question and some others had no clear strategy for finding a solution. One error was to find the time to travel 2m and then use this time to show that they travelled 2 m. Some candidates found the speed, when the rope became slack, but could not then progress to deal with the motion that took place after this had happened.

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Q9.

While there were some good proofs of the formula for the range, there were also a number of confused approaches. Several of these were based on the formula as $v^2 = u^2 + 2as$. However, those who found the time of flight first often made quite good progress. Some candidates misquoted the double angle formula. Another area of difficulty was in the solution of the quadratic equation to find the time of flight.

In part (b), almost all of the candidates gave the answer 45° , but very few were able to give a convincing argument.

Part (c) was generally done very well. Some candidates tried to do this without using the results from parts (a) and (b). Many of these candidates were unsuccessful. A few candidates did not give their answer for the range to the nearest metre as instructed in the question.

Q10.

The candidates found this question very demanding and there were very few complete solutions. Many candidates obtained marks of 0 or 1 on this question. While a good number of candidates could find the vector $\vec{AB}$, many candidates added the two vectors rather than subtracting. In the other parts of the question many candidates simply did not recognise how to proceed. Several tried to find the magnitudes of the vectors or use equations of the form as $v^2 = u^2 + 2as$. Some candidates made reasonable attempts at parts (c) and (d), but using an incorrect velocity from part (b). The candidates who produced good solutions were clearly very confident when working with vectors.

Q11.

This question was often done well. Parts (a) and (b) were answered correctly by a large proportion of the candidates, but part (c) caused more difficulties. The most common incorrect responses were 1225 N, caused by a sign error in the equation of motion, or 1625 N, obtained by ignoring the resistance force.

Q12.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q13.

Part (a) was generally done well. Some candidates made arithmetic errors and a small number misquoted the constant acceleration equation. The force diagrams were drawn very well. Most of the candidates found it easy to calculate the magnitude of the friction force, although a small number took the reaction force as being equal to the weight. Many candidates had difficulty finding the tension in the rope.

Q14.

The candidates generally did very well with this question. Apart from a few minor errors in earlier parts, the only problems arose in part (c)(ii), where a number of candidates found the difference between 768 and 450, instead of the sum.

Q15.

The candidates had very few problems with this question, with a high proportion of completely correct solutions.

Q16.

There were a number of good solutions to part (a). Some candidates did make sign errors when setting up the equation for the vertical motion. Some candidates broke the motion down into stages, but those that did often made small errors. Part (b) was done very well, with some candidates using the time given in part (a) to enable them to find the range. Part (c) proved to be very demanding. Some candidates calculated the vertical component of the velocity when the car hit the sea, but did not bring the horizontal component into their solutions. There were only very few complete solutions.

Q17.

Part (a) was generally done very well. Very few candidates did not form two equations of motion as requested. A few had problems dealing with the friction force. Part (b) proved to be more difficult for some candidates, who dropped the negative sign and simply took the acceleration to be 1.225. Many of these candidates then just dropped a negative sign in the middle of their working. Part (c) was done well by some candidates, but again the negative acceleration caused problems for some.

Q18.

This question was quite challenging for the candidates and there were very few good solutions, although most candidates did pick up marks from one or two parts of the question. Part (a) was done by a fair number of candidates. Errors included treating the position at $t = 10$ as a velocity and not using this position vector at all. There were some good attempts at part (b) with candidates using their acceleration from part (a). Part (c) caused problems for many candidates. It was not uncommon to see answers where the time was given as a vector. Some candidates only considered one of the components. There was clearly a number of candidates who could write down a vector equation, but could not solve it to find T . In part (c)(ii), several candidates only found the velocity and did not go on to find the speed.



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